FREEDOM INTERNATIONAL SCHOOL **CHAPTER – d and f BLOCK ELEMENTS** **Question and Answer (Type – Give reasons)**	
1.	Transition metals and their compounds are generally found to be good catalysts. REASON: The catalytic properties of the transition elements are due to the presence of unpaired electrons in their incomplete d-orbitals and variable oxidation states.
2.	Metal-metal bonding is more frequent for the 4d and the 5d series of transition metals than that for the 3d series. REASON: Metal-metal bonding is more frequent for the 4d and the 5d series of transition metals than that for the 3d series as these have their electrons of outer most shell at greater distance from the nucleus, as compared to atoms of 3d transition metals.
3.	Transition metals and their compounds generally exhibit a paramagnetic behaviour. REASON: Because of presence of unpaired electrons in their d-subshell in atomic and ionic state.
4.	Generally, there is an increase in density of elements from titanium (Z = 22) to copper (Z = 29) in the first series of transition elements. REASON: From titanium to copper the atomic size of elements decreases and mass increases as a result of which density increases.
5.	Transition elements generally form coloured compounds. REASON:

	Because of presence of unpaired d electrons, which undergoes d-d transition by absorption of energy from visible region and then the emitted light shows complementary colours. This is how transition elements form coloured compounds.
6.	Zinc is not regarded as a transition element. REASON: Zinc in its common oxidation state of +2 has completely filled d-orbitals. Hence considered as non-transition element.
7.	Copper (I) ion is not known in aqueous solution. REASON: $Cu^{2+}(aq)$ is much more stable than $Cu^{+}(aq)$. This is because although second ionization enthalpy of copper is large but Δ_{hyd} (hydration enthalpy) for $Cu^{2+}(aq)$ is much more negative than that for $Cu^{+}(aq)$ and hence it more than compensates for the second ionization enthalpy of copper. Therefore, many copper (I) compounds are unstable in aqueous solution and undergo disproportionation as follows: $2Cu^{+} \rightarrow Cu^{2+} + Cu$
8.	Manganese exhibits the highest oxidation state of +7 among the 3d series of transition elements. REASON: Because Mn - $3d^5 4s^2$ it has maximum no. of electrons to lose or share. So, it can exhibit oxidation state of +2 to +7 which is maximum number.
9.	Highest oxidation State of manganese in fluorides is +4 (MnF_4) but highest oxidation state in oxides is +7 (Mn_2O_7) REASON: Mn shows the highest o.s of +7 with oxygen because it can form p-pi-d-pi multiple bonds using 2p orbital of oxygen and 3d orbital of Mn. On the other hand, with fluorine, the highest state of +4 is seen because it can form only single bond.
10.	Cr^{2+} is reducing in nature while with the same d-orbital

	configuration (d^4) Mn^{3+} is an oxidising agent. REASON: Cr^{2+} has the configuration $3d^4$ which easily changes to d^3 due to stable half-filled t_2g orbitals. Therefore Cr^{2+} is reducing agent. While Mn^{2+} has stable half-filled d^5 configuration. Hence Mn^{3+} easily changes to Mn^{2+} and acts as oxidising agent.
11.	In a transition series of metals, the metal which exhibits the greatest number of oxidation states occurs in the middle of the series. REASON: Due to presence of more unpaired electrons and use of all 4s and 3d electrons in the middle of series.
12.	The enthalpies of atomization of transition metals are quite high. REASON: In transition elements, there are large number of unpaired electrons in their atoms, thus they have a stronger inter atomic interaction and thereby stronger bonding between the atoms. Due to this they have high enthalpies of atomization.
13.	The ionization enthalpies (first and second) in the first series of the transition elements are found to vary irregularly. REASON: Due to irregularities in the electronic configuration there is irregularities in the enthalpies of atomisation. Hence there is irregular variation in I.E.
14.	Both O_2 and F_2 stabilize high oxidation states of transition metals but the ability of oxygen to do so exceeds that of fluorine. REASON: The ability of O_2 to stabilize higher oxidation states exceeds that of fluorine because oxygen can form multiple bonds with

	metals.
15.	Copper atom has completely filled d orbitals ($3d^{10}$) in its ground state, yet it is regarded as a transition element. REASON: Copper atom has completely filled d orbitals ($3d^{10}$) in its ground state, yet it is regarded as a transition element due to incompletely filled d-orbital in its ionic states i.e. Cu^{2+} ($3d^9$).
16.	Cr^{2+} is a stronger reducing agent than Fe^{2+} in aqueous solutions. REASON: The highest oxidation state for Cr is +6, therefore it can loose 3 more electrons, whereas Fe needs to loose only 1 electron to achieve its highest oxidation state of +3. Thus, Cr^{3+} is more reducing than Fe^{2+}.
17.	Silver atom has completely filled d-orbitals ($4d^{10}$) in its ground state, yet it is regarded as a transition element. REASON: Because silver has incomplete d-orbital ($4d^9$) in its +2 oxidation state, hence it is a transition element.
18.	E^0 value for Mn^{3+} /Mn^{2+} couple is much more positive than Cr^{3+}/Cr^{2+}. REASON: The large positive E° value for Mn^{3+}/Mn^{2+} shows that Mn^{2+} is much more stable than Mn^{+3} due to stable half filled configuration ($3d^5$). Therefore the 3rd ionisation energy of Mn will be very high and Mn^{3+} is unstable and can be easily reduced to Mn^{2+}. E° value for Cr^{3+} | Cr^{2+} is positive but small i.e. Cr^{3+} can also be reduced to Cr^{2+} but less easily. Thus Cr^{3+} is more stable than Mn^{3+}.
19.	The metallic radii of the third (5d) series of transition elements are virtually the same as those of the corresponding members of the second series.

	REASON: Due to lanthanoid contraction in second series after lanthanum, the atomic radii of elements of second and third series become almost same and hence show similarities in properties.
20.	The E° value for the Mn^{3+}/Mn^{2+} couple is much more positive than that for Cr^{3+}/Cr^{2+} couple or Fe^{2+}/Fe^{2+} couple. REASON: The electronic configuration of Mn^{2+} ion is more symmetrical as compared to that of Cr^{2+} ion. So 3rd ionisation potential of Mn^{2+} is much higher. As a result E° value of Mn^{3+}/Mn^{2+} couple is much more positive than for Cr^{3+}/Cr^{2+} couple.
21.	Mn^{2+} is much more resistant than Fe^{2+} towards oxidation. REASON: The 3d orbital in Mn^{2+} is half-filled and is more stable compared to Fe^{2+} has 6 electrons in the 3d orbital. Mn^{2+} prefer to lose an electron or get oxidised whereas Fe^{2+} will readily loose one electron or get oxidised. Therefore, Mn^{2+} is much more resistant than Fe^{2+} towards oxidation.
22.	Among lanthanoids, Ln (III) compounds are predominant. However, occasionally in solutions or in solid compounds, + 2 and + 4 ions are also obtained. REASON: Some lanthanoid show +2 and +4 oxidation states in ionic solutions or solid components due to the extra stability that arises because of empty half-filled or fully filled 4f-subshell. This is why occasionally in solution or in solid compounds, +2 and +4 ions are also obtained. even though Ln (III) compounds are predominant.
23.	The $E°_{M^{2+}}/M$ for copper is positive (0.34 V) REASON: $E°_{M^{2+}}/M$ for any metal is related to the sum of enthalpy of

	atomisation, ionization and hydration enthalpy. Copper has high enthalpy of atomisation and low enthalpy of hydration. Hence $E°_{Cu^{2+}/Cu}$ is positive. The high energy required to transform $Cu_{(s)}$ to $Cu^{2+}_{(aq)}$ is not balanced by its hydration enthalpy.
24.	There is hardly any increase in atomic size with increasing atomic numbers in a series of transition metals. REASON: Along a transition series, the nuclear charge increases which tends to decrease the size but the addition of electrons in the penultimate d-subshell increases the screening effect which counter balances the effect of increased nuclear charge. Thus, atomic radius does not change much.
25.	Zr (Z = 40) and Hf (Z = 72) have almost identical radii. REASON: Zr and Hf have almost identical radii due to lanthanoid contraction which is due to weak shielding of d-electrons.
26.	MnO is basic while Mn_2O_7 is acidic. REASON: In low oxidation state of the metal, some of the valence electrons of the metal atom are not involved in bonding. Hence, it can donate electrons and behave as a base. On the other hand, in higher oxidation state of the metal, valence electrons are involved in bonding and are not available. Instead, effective nuclear charge is high and hence it can accept electrons and behave as an acid. (MnO (+4) and Mn_2O_7 (+7)) The oxide in the lower oxidation state of a metal is basic and in the higher oxidation state of the metal it is acidic.
27.	The $E°_{Mn^{2+}/Mn}$ value for manganese is much more than expected from the trend for other elements in the series. REASON:

	The value of E° for Mn is more negative than expected from the general trend due to greater stability of half filled d-subshell (d^5) in Mn^{2+}.
28.	The oxidising power of oxoanions are in the order $VO_2^+ < Cr_2O_7^{2-} < MnO_4^-$ REASON: The ions in which the central metal atom is present in the highest oxidation state will have the highest oxidising power. In VO+2 V is present in the +5 oxidation state, while in $Cr_2O_7^{2-}$ ion, Cr is present in the +6 oxidation state. Similarly, in the MnO_4^- ion, Mn is present in the +7 oxidation state. Thus, as the oxidation state of the central metal atom increases in the order V < Cr < Mn, therefore the oxidising power of the oxoanions increases in the order $VO_2^+ < Cr_2O_7^{2-} < MnO_4^-$
29.	Cr^{2+} is a stronger reducing agent than Fe^{2+}. REASON: Cr^{2+} is a stronger reducing agent than Fe^{2+} because $E°_{Cr^{3+}/Cr^{2+}}$ is negative (-0.41 V) whereas $E°_{Fe^{3+}/Fe^{2+}}$ is positive (0.77 V). Thus Cr^{2+} is easily oxidised to Cr^{3+} but Fe^{2+} cannot be easily oxidized to Fe^{3+}.
30.	The lowest oxide of transition metal is basic whereas the highest oxide is amphoteric or acidic. REASON: The lowest oxide of transition metal is basic because of low oxidation state some of valence electrons are not involved in bonding and acts as base by donating electron. However, in higher oxides due to high oxidation state, it cannot donate electrons but can accept electrons due to high effective nuclear charge. Hence, they are acidic in nature.
31.	Unlike Cr^{3+}, Mn^{2+}, Fe^{3+} and the subsequent other M^{2+} ions of the 3d series, the 4d and the 5d series metals generally do not form stable cationic species.

	REASON: 4d and 5d series metals generally have much higher enthalpies of atomization, leading to stronger metal-metal bonds. In 4d and 5d series, higher oxidation states are more stable because the energy gap between the energy levels is small hence favouring covalent nature in their compounds